A PLAIN-LANGUAGE WHITE PAPER

ISTOE

A universal root cause analysis engine.

ISTOE stands for the Incentive Structures Theory of Everything. That name sounds grand, so here is the honest version of what it means: **"everything" means at every scale** — not that one idea explains every phenomenon.

The same patterns of incentives show up in a single person, a family, a workplace, a company, and a whole country. ISTOE is a way of spotting those patterns, so you can find the real cause of a problem instead of arguing about the symptom.

> Stop asking, "What is wrong with these people?"
> Start asking, "What is this system rewarding?"

Think of it the way you would think of a flowchart or an organization chart. An org chart does not explain a whole company — it makes the authority structure visible, so you can act on it. A flowchart does not explain a whole process — it makes the steps visible. An ISTOE map does the same thing for rewards: it draws who wants what, who is in conflict with whom, and which rules are quietly pushing everyone's behavior.

Once you draw that map, something useful happens. A problem that looked small and personal often turns out to be large and structural. A problem that looked like a character flaw often turns out to be a predictable response to the rules. And a problem that looked impossible often turns out to have a specific rule, at a specific place, that could be changed.

This paper explains the framework in ordinary language: the idea, the patterns to look for, a step-by-step method anyone can use, and — just as important — what ISTOE does *not* claim.

Start with a different question

Most arguments about problems start downstream, at the visible end. Young people are called lazy. Workers are called unmotivated. Customers are called fickle. Executives are called greedy. Sometimes those descriptions even feel satisfying — but they rarely explain anything, and they almost never point to a fix.

ISTOE starts upstream instead. People respond, mostly predictably, to the situation they are in: what it costs them, what it pays them, what it punishes, and what it makes easy or hard. Change the situation and the "character" of large groups of people appears to change with it — because the behavior was never mainly about character in the first place.

A familiar example

Suppose a generation of young adults works less, saves less, and moves out later than their parents did. The downstream story is a story about attitude. The upstream story asks different questions. What does an entry-level wage actually buy now, compared with then? What do housing and credentials cost? If honest effort no longer reliably converts into stability, stepping back is not a mystery. It is arithmetic.

> **The core premise:** human behavior, organizational outcomes, and societal trends are driven overwhelmingly by upstream incentive structures and rule-sets — not by individual willpower, personal virtue, or moral failing. Systems shape behavior far more reliably than character does.

That premise does not excuse anyone from anything. It does something more practical: it tells you where the leverage is. You cannot redesign a generation's character. You can redesign a pay structure, an application process, a tax rule, or an app's notifications — and behavior will follow the redesign.

It also changes the emotional temperature. When you strip the moral and emotional charge off a scenario — temporarily, as a working method — the structure becomes visible. Stakeholders stop being heroes and villains and become people responding to pressures you can actually map.

Four patterns explain a lot of what you see

These are the framework's four diagnostic pillars, stated plainly. Treat them as four questions to ask about any problem.

1. What does effort pay?

Everyone spends a finite budget of time, attention, and energy. When effort reliably pays off, people engage. When the return collapses — high costs, erratic rewards — people disengage. That is energy conservation, not laziness.

2. Who is getting blamed?

Public argument tends to blame individuals for outcomes a system produced. Blaming people is comfortable for the system: if the problem is their character, the rules never have to change.

3. Who does the interface punish?

Systems are usually built for one "default" kind of person. People whose minds or skills do not match that default get penalized for failing paperwork, formats, and compliance hurdles — not for lacking ability.

4. What rule changed?

Big shifts in corporate and economic behavior usually follow a rule change. In 1982, for example, a new securities rule made it much safer for companies to buy back their own stock — and buybacks became routine for decades afterward.

Notice what the four patterns have in common. None of them requires anyone to be stupid, lazy, or evil. Each one describes a situation in which ordinary people, responding sensibly to the costs and rewards in front of them, produce an outcome that then gets blamed on their character.

That is the diagnostic move in one sentence: **before you judge the output, audit what the rule-set is paying for.**

Three instincts built into almost every system

Underneath the four patterns sit three tendencies that recur at every scale, from a single habit to a national policy. They are the load-bearing part of the mental map.

1 Reward hacking

Any system that chases a reward will find the shortcut between the reward and the real goal. A student crams for the test instead of learning the subject. A company hits its sales target by discounting away its profit. An app user scrolls for the next small hit instead of the rest they actually wanted. The measure got satisfied; the goal did not. This is not just a problem in artificial intelligence, where the term comes from — it is an old evolutionary pattern in humans.

2 Maximum output, minimum input

People instinctively look for the most return for the least effort. That is not a moral flaw; for most of human history it was survival. Conserving calories and energy kept you alive when food was scarce. The instinct misfires in a world of abundance, where the cheapest, easiest reward — the snack, the scroll, the shortcut — is available instantly and endlessly, and the survival loop gets hijacked by stimuli designed to trigger it.

3 A bias toward continuity

Systems and the people in them strongly prefer keeping things as they are over adapting or changing. Change costs energy, carries risk, and requires many people to move at once. So the current arrangement tends to persist even when a better one is visible to everyone — which is why redesigning a system is usually harder than diagnosing it.

> Put the three together and many puzzles dissolve. People are not mysteriously irrational. They are conserving energy, chasing the rewards actually on offer, and defending the arrangement they already know — inside rules somebody else set.

How to use ISTOE, step by step

0 **Set the moral charge aside.** Not forever, and not because right and wrong do not matter — but because outrage makes structure hard to see. Examine the situation logically first; judge it after you understand it.

1 **Map everyone involved.** List every stakeholder — including the quiet ones and the ones nobody speaks for. For each one, ask: what are they rewarded for, punished for, and afraid of?

2 **Follow each incentive to the end.** Do not stop at the surface want. “More money” is usually a means to security, freedom, status, or peace of mind. Stopping early — truncating the map — is the most common way an analysis goes wrong, because it makes solvable conflicts look like permanent standoffs.

3 **Name the pattern.** Is this reward hacking? A collapsed return on effort? A mismatched interface? A continuity lock-in? Naming the pattern tells you which known failure modes to check.

4 **Treat the system itself as changeable.** The current rules are not laws of nature. Ask what would happen if the rule, price, default, or interface were different.

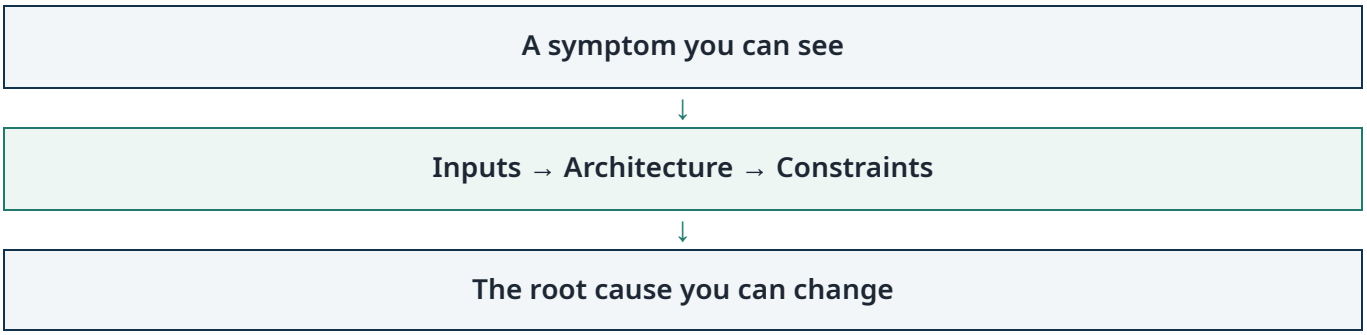
5 **Redesign so the good behavior is the easy behavior.** The goal is a rule-set where doing the desired thing is also the locally sensible thing for each person in it — so cooperation becomes the default, not the exception.

Four lenses for a first look at any system

ISTOE generalizes a troubleshooter's toolkit. The same root-cause analysis used on a computer network, a game engine, or a piece of software works on a workplace or on your own habits — because in each case you are reading a system through the same four lenses.

Lens	The question it asks
Inputs	What is driving the behavior or the error? What pressures, rewards, and collisions are feeding into this system?
Architecture	What are the pathways? How do the parts connect, and where does the signal — money, attention, approval, information — actually travel?
Constraints	What are the hard limits? Time, energy, money, attention, and processing capacity are finite, in people just as in machines.
Output	What is the final state? Burnout, churn, debt, anxiety, growth — the output is the symptom you noticed first, and usually the last thing to blame.

Worked this way, the same logical toolkit debugs a script, analyzes a corporate structure, or maps a person's emotional architecture. That is the payoff of the approach: reality stops being a chaotic black box and becomes a transparent, editable schematic — something you can draw, share, argue about, and revise.



The diagram above is the whole method in miniature: start at the output, work backward through the system, and stop when you reach something that can actually be redesigned.

What people actually want, underneath

Follow any stakeholder's incentives far enough and the map bottoms out in the same human territory. People seek experiences that feel good and avoid ones that feel bad, across nested loops running at different speeds: the minute-to-minute loop of a pleasant conversation, the hour-to-hour loop of making progress, the day-to-day loop of a life that feels like it is going somewhere. It is structured much like a video game, which is one reason game design is such a clear teacher of this framework — in a game, the loops are written down where everyone can see them.

Three of those loops matter most, and they have names: **autonomy** (being in charge of your own choices), **competence** (being good at something and getting better), and **relatedness** (being connected to other people). This is why a pay raise that destroys someone's autonomy can still leave them miserable, and why "just pay them more" so often fails as a fix.

Wanting is not the same as liking

Research on the brain's reward systems adds an important refinement: the drive to *want* something and the pleasure of *liking* it are separate systems, and they can come apart. In addiction, wanting keeps growing while liking stays flat or falls. That gap is exactly where exploitation lives.

> **Genuine satisfaction vs. exploitation.** A well-designed system satisfies people's loops for real: the notification that actually connects you to a friend, the game that actually leaves you refreshed. An exploitative one triggers the loop without delivering the goods: the endless scroll that activates wanting and delivers nothing. ISTOE's design test is simple to state — does this system satisfy the loop, or does it just stimulate it?

The point of all this: raise the floor

A framework for redesigning systems needs a goal, stated out loud, so that whoever draws the map is bound by it too. ISTOE's goal is to **raise the floor** — to lift the baseline of human existence, measured by quality of life, lifespan, and healthspan.

Why the floor, and not the average? Because an average can look healthy while hiding people in misery underneath it. A system's score, in this framework, is its minimum: how well the worst-off person in it is actually doing. That choice also keeps the goal honest. A system cannot claim success by stimulating people's reward loops while their health collapses — lifespan and healthspan are part of the score, so that shortcut is ruled out by construction.

The diagnostic and prescriptive halves mirror each other. The diagnostic half is largely a story of floors collapsing — effort stops paying, the baseline of a decent life becomes unaffordable, and disengagement follows as a rational response. The prescriptive half is the reverse: find the rule that is holding the floor down, and redesign it.

Two quick examples

A workplace. A team keeps missing deadlines and management calls it an attitude problem. The map shows something else: people are rewarded for starting visible projects, not finishing invisible ones; asking for help is quietly punished; and the effort-to-recognition return is near zero. Redesign the rewards, and the "attitude" changes.

A phone habit. Someone stays up too late scrolling and calls it a willpower problem. The map shows an app engineered to hack the wanting loop, notifications timed to moments of low energy, and no stopping rule anywhere in the design. The fix is architectural — change the defaults, the friction, and the triggers — not a sterner lecture to the user at midnight.

What ISTOE does not claim

A tool you can trust is a tool whose limits are printed on the box. ISTOE is a diagnostic framework and an index of incentive patterns — a lens and a method — not a complete theory of human nature and not a machine that outputs correct answers.

- **Motivation is not one thing.** Wanting and liking come apart, and meaning, duty, and identity move people in ways a simple pleasure account does not capture. The framework starts with incentives; it does not end there.
- **Incentives change people, not just behavior.** Adding a fine or a bonus can rewrite a social norm permanently — a famous daycare study found that fining late pickups made them *more* common, and lateness stayed high even after the fine was removed.
- **Perfect alignment is impossible.** Established results in social choice and mechanism design prove that no procedure can satisfy everyone, honestly and completely, at once. Every real redesign must say what it is trading off.
- **There is no neutral mapper.** Whoever draws the map has incentives too. That is why the goal — raise the floor — is stated explicitly, and why maps should be drawn openly, with the stakeholders able to argue with them.
- **The floor is a choice, not a discovery.** Choosing to judge a system by its worst-off is a moral commitment. ISTOE states it openly rather than pretending it was derived from the data.

The one-sentence version

> Strip away the blame, map who is rewarded for what, follow it to the root — then change the rules so the floor rises and the good behavior becomes the default.

That is ISTOE: the troubleshooter's habit of mind, generalized to everything at every scale — a universal root cause analysis engine for finding the real cause of a problem, and a fairer place to fix it.

ISTOE was developed by Karl Burniston. This layperson's edition presents the framework in his own terms, in plain language, including the limits recorded for its v1.0 direction.